

$$A. e^{-2t} u(t) + e^{-3t} u(t)$$

$$\mathcal{L}\{e^{-2t} u(t)\} + \mathcal{L}\{e^{-3t} u(t)\}$$

$$= \int_0^{\infty} e^{-2t} e^{-st} dt + \int_0^{\infty} e^{-3t} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+2)t} dt + \int_0^{\infty} e^{-(s+3)t} dt$$

$$= \left[ -\frac{e^{-(s+2)t}}{s+2} \right]_0^{\infty} + \left[ -\frac{e^{-(s+3)t}}{s+3} \right]_0^{\infty}$$

$$\text{ROC: } s > -2 \quad \text{and} \quad s > -3$$

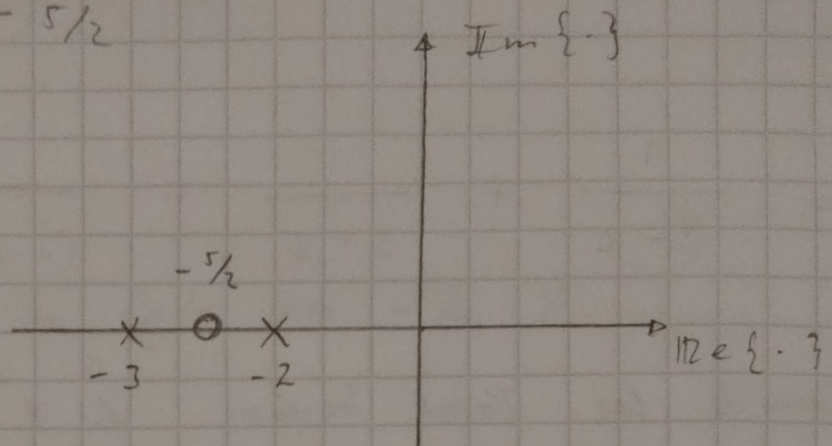
$$\frac{1}{s+2} + \frac{1}{s+3} = \frac{2s+5}{(s+2)(s+3)} \rightarrow \text{zeros}$$

$$\rightarrow \text{Poles}$$

donde los Polos serán en  $-2$  y  $-3$

y los ceros en  $-5/2$

2/



$$13. e^{2t} u(t) + e^{-3t} u(t)$$

$$\mathcal{L}\{e^{2t} u(t)\} + \mathcal{L}\{e^{-3t} u(t)\}$$

$$\int_0^{\infty} e^{2t} \cdot e^{-st} dt + \int_0^{\infty} e^{-3t} \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s-2)t} dt + \int_0^{\infty} e^{-(s+3)t} dt$$

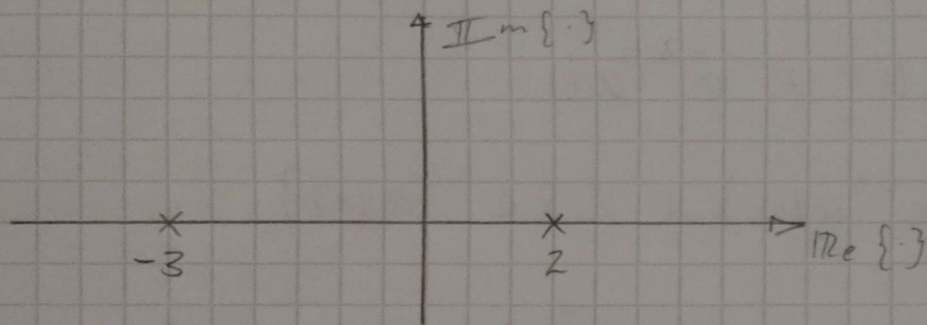
$$= \left[ -\frac{e^{-(s-2)t}}{s-2} \right]_0^{\infty} + \left[ -\frac{e^{-(s+3)t}}{s+3} \right]_0^{\infty}$$



$$\text{ROC: } s > 0 \quad \text{y} \quad s < -3$$

$$= \frac{1}{s-2} + \frac{1}{s+3} = \frac{s}{(s-2)(s-3)} \rightarrow \text{Sin polos}$$

Donde los Polos serán: 2 y -3



$$c. e^{-a|t|}$$

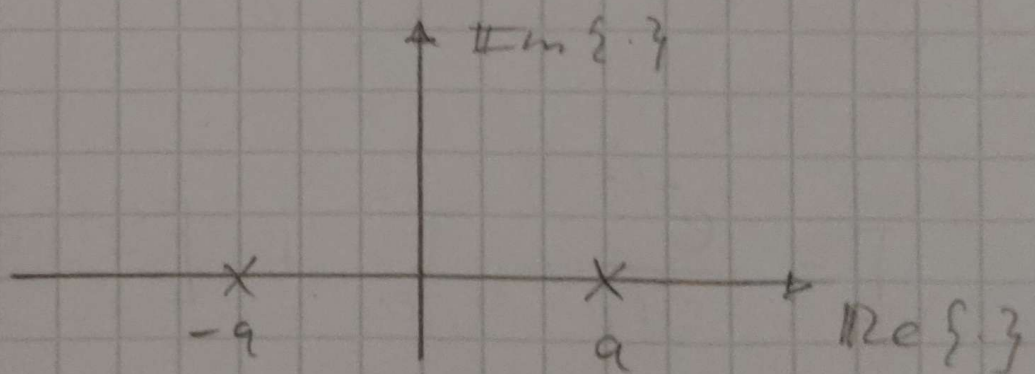
$$\begin{aligned} \mathcal{L}\{e^{-a|t|}\} &= \int_{-\infty}^0 e^{-a(-t)} e^{-st} dt + \int_0^{\infty} e^{-at} e^{-st} dt \\ &= \int_{-\infty}^0 e^{-(s-a)t} dt + \int_0^{\infty} e^{-(s+a)t} dt \end{aligned}$$

$$= \left[ \frac{-e^{-(s-a)t}}{s-a} \right]_{-\infty}^{\infty} + \left[ \frac{-e^{-(s+a)t}}{s+a} \right]_{0}^{\infty}$$

$$\text{ROC: } s > -a \quad \text{y} \quad s < a$$

$$= \frac{1}{s-a} + \frac{1}{s+a} = \frac{2a}{s^2 + a^2}$$

Donde solo hay polos en  $a$  y  $-a$ :





$$D. e^{-2t} [u(t) - u(t-5)] ; t \in [0, 5]$$

$$\mathcal{L}\{e^{-2t} [u(t) - u(t-5)]\}$$

$$= \int_0^5 e^{-2t} \cdot e^{-st} dt$$

$$= \left[ -\frac{e^{-(2+s)t}}{2+s} \right]_0^5$$

$$= \frac{1 - e^{-(5s+10)}}{s+2} \quad \begin{array}{l} \rightarrow \text{zeros} \\ \rightarrow \text{Poles} \end{array}$$

$$\text{ROC} \rightarrow s > -2$$

$$1 - e^{-(5s+10)} = 0$$

$$\ln(1) = -5s + 10$$

5]

$$0 = -5s + 10$$

$$s = 2$$

P/

Donde hay un Polo en  $-2$  y  
un cero en  $2$ :

